Measures of Center & Spread

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Mean and median**

Find the mean and median of the data set: $4, 7, 7, 9, 13$.

Mean $= \frac{4+7+7+9+13}{5} = \frac{40}{5} = 8$. Median (middle value of the ordered list): 7.

2 **[Foundational]** • **Target: Mean and median**

A sixth data point of 40 is added to the set $4, 7, 7, 9, 13$. Recompute the mean and median, and explain which measure changed more.

New mean $= \frac{40+40}{6} = \frac{80}{6} \approx 13.33$. New median (ordered $4, 7, 7, 9, 13, 40$, average of the two middle values 7 and 9): 8. The mean changed drastically (from 8 to 13.33) while the median barely moved (from 7 to 8), since the mean is sensitive to the extreme outlier 40 while the median is resistant to it.

3 **[Foundational]** • **Target: Range, IQR, and quartiles**

Find the five-number summary (min, Q_1, median, Q_3, max) of the ordered data: $2, 5, 6, 9, 11, 14, 15, 18, 20$.

Min $= 2$. Median (middle of 9 values): 11. Lower half $2, 5, 6, 9$: $Q_1 = \frac{5+6}{2} = 5.5$. Upper half $14, 15, 18, 20$: $Q_3 = \frac{15+18}{2} = 16.5$. Max $= 20$.

4 **[Foundational]** • **Target: Range, IQR, and quartiles**

Using the five-number summary from the previous problem, find the IQR and determine whether 20 would be flagged as an outlier using the $1.5 \times$ IQR rule.

IQR $= Q_3 - Q_1 = 16.5 - 5.5 = 11$. Upper fence $= Q_3 + 1.5(11) = 16.5 + 16.5 = 33$ (using the $1.5 \times$ IQR rule). Since $20 < 33$, it is not flagged as an outlier.

5 **[Foundational]** • **Target: Standard deviation and variance**

Find the standard deviation of the data set $2, 4, 6, 8$ (treat as a population).

Mean $= 5$. Deviations: $-3, -1, 1, 3$. Squared deviations: $9, 1, 1, 9$, sum $= 20$. Variance $= \frac{20}{4} = 5$. Standard deviation $= \sqrt{5} \approx 2.24$.

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6 [Foundational] • Target: Standard deviation and variance

Explain why the standard deviation is always non-negative, and why it equals 0 only in one specific case.

The standard deviation is the square root of an average of squared deviations, and squares are always non-negative, so their average and its square root are also non-negative. It equals 0 only when every data value is identical (no spread at all), since that is the only case where every deviation from the mean is 0.

7 [Foundational] • Target: Visualizing spread with a normal-shaped distribution

The graph shows a bell-shaped distribution of exam scores with mean 75 and standard deviation 8.

a Estimate the range of scores that fall within one standard deviation of the mean. One standard deviation: 75 ± 8, so from 67 to 83.

b Estimate the range of scores that fall within two standard deviations of the mean. Two standard deviations: 75 ± 16, so from 59 to 91.

(a) One standard deviation: 75 ± 8, so from 67 to 83.; (b) Two standard deviations: 75 ± 16, so from 59 to 91.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Comparing distributions using measures of spread

Class A has a mean test score of 80 with a standard deviation of 4. Class B has a mean of 80 with a standard deviation of 12. Compare the two classes' consistency, and explain what a small vs. large standard deviation indicates.

Both classes have the same average performance (mean 80), but Class A's scores are tightly clustered around the mean (small spread, standard deviation 4), meaning most students scored similarly, while Class B's scores are much more spread out (standard deviation 12), meaning there is far more variability — some students scored well above or below the average.

9 [Intermediate] • Target: Resistant vs. non-resistant statistics

Explain why the median and IQR are described as "resistant" measures, while the mean and standard deviation are not, using the outlier example from earlier in this worksheet as a reference.

A resistant measure is one that is not strongly affected by extreme values or outliers. The median and IQR depend only on the position (ranking) of data values, not their exact size, so a single extreme value has little effect on them. The mean and standard deviation, in contrast, use every data value's actual size in their calculation, so one extreme outlier (like the 40 added earlier) can shift them substantially.

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10 **[Intermediate]** • **Target: Choosing appropriate summary statistics**

A data set of home prices in a neighborhood is strongly skewed right due to a few very expensive mansions. Recommend which pair of summary statistics (mean/standard deviation or median/IQR) should be reported, and justify your choice.

The median and IQR should be reported, since they are resistant to the effect of the few extremely expensive mansions. Reporting the mean would give a misleadingly high sense of a "typical" home price, since the mean is pulled upward by the small number of very high values.

11 **[Intermediate]** • **Target: Choosing appropriate summary statistics**

A quality-control process measures the diameter of manufactured bolts, which are expected to be very consistent (low variability) and roughly symmetric. Recommend which pair of summary statistics is more appropriate here, and explain why.

The mean and standard deviation are appropriate here, since the data is roughly symmetric with no extreme outliers expected — in this case, the mean and standard deviation use all the data efficiently and are the standard choice when a distribution is not skewed.

12 **[Intermediate]** • **Target: Computing summary statistics from a frequency table**

A survey of families records number of children per household: 0 children (8 families), 1 child (15 families), 2 children (20 families), 3 children (7 families). Find the mean number of children per household.

Total families: $8 + 15 + 20 + 7 = 50$. Total children:
$0(8) + 1(15) + 2(20) + 3(7) = 0 + 15 + 40 + 21 = 76$. Mean $= \frac{76}{50} = 1.52$ children per household.

13 **[Intermediate]** • **Target: Computing summary statistics from a frequency table**

A small business records weekly sales (in thousands): $10, 12, 12, 14, 17$. Find the mean, and then find the variance and standard deviation (treat as a population).

Mean $= \frac{10 + 12 + 12 + 14 + 17}{5} = \frac{65}{5} = 13$. Deviations: $-3, -1, -1, 1, 4$. Squared: $9, 1, 1, 1, 16$, sum $= 28$. Variance $= \frac{28}{5} = 5.6$. Standard deviation $= \sqrt{5.6} \approx 2.37$.

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14 **[Intermediate]** • **Target: Computing summary statistics from a frequency table**

Two data sets both represent quiz scores out of 20 for two different classes of the same size, and both have the same mean of 15. Class A has a standard deviation of 1.2, while Class B has a standard deviation of 4.5. Without doing any further calculation, describe what a dotplot of each class's scores would likely look like, and explain how you can tell just from the two standard deviations.

Class A's dotplot would show scores tightly clustered close to 15 (most students scoring similarly, in a narrow band), since a small standard deviation of 1.2 indicates little spread around the mean. Class B's dotplot would show scores much more spread out around 15, with a substantial number of students scoring noticeably above or below the mean, since a standard deviation of 4.5 indicates far greater variability.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Computing summary statistics from a frequency table** • **Cross-Lesson: 3.01 Probability Rules & Conditional Probability**

A coach records the vertical jump heights (in inches) of 7 athletes: $22, 24, 25, 26, 26, 28, 41$. Explain why the value 41 would likely be flagged as an outlier, and describe, in general terms, the two-step process for using the $1.5 \times$ IQR rule to formally check whether a value is an outlier.

The value 41 is far larger than the rest of the data (which clusters between 22 and 28), so it looks like a clear outlier by visual inspection. Formally, the $1.5 \times$ IQR rule works in two steps: first compute the IQR ($Q_3 - Q_1$) from the five-number summary, then compute the lower fence ($Q_1 - 1.5 \times$ IQR) and upper fence ($Q_3 + 1.5 \times$ IQR); any value falling below the lower fence or above the upper fence is classified as an outlier.

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16 **[Challenging]** • **Target: Computing summary statistics from a frequency table** • **Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions**

A teacher records the number of minutes 8 students spent on homework last night: $30, 45, 50, 55, 60, 60, 65, 200$. The value 200 comes from a student who mistakenly reported total weekly homework time instead of just last night. Explain, step by step, how this single incorrect data entry would affect the reported mean versus the reported median, and explain which measure a teacher should trust more when summarizing typical homework time for this data set, given the known data-entry error.

The mean would be pulled dramatically upward by the erroneous 200 value, since the mean uses every data point's exact size — it would suggest students spend far more time on homework than they actually do. The median, which only depends on the middle-ranked value(s), would be far less affected by this single extreme (and erroneous) value. Given the known data-entry error, the teacher should trust the median as a more accurate reflection of typical homework time, since it resists distortion from this kind of outlier.

17 **[Challenging]** • **Target: Comparing distributions using measures of spread** • **Cross-Lesson: 3.03 The Normal Distribution**

A machine dispenses a liquid into containers. The amount of liquid dispensed, in milliliters, follows a continuous uniform distribution over the interval $[95, 105]$. A quality control inspector randomly selects 5 containers and measures the amount of liquid in each. Let X_i be the amount of liquid in the ith container, for $i = 1, 2, 3, 4, 5$. The company has implemented a new calibration process that, on average, increases the mean amount of liquid dispensed by δ milliliters, but also increases the variance by σ^2. The inspector is specifically looking for evidence that the calibration process has *decreased* the variability of the dispensed liquid while maintaining a satisfactory average fill. If the current calibration process yields a mean fill of $\mu_0 = 100$ ml and a standard deviation of $\sigma_0 = 2$ ml, and the new process is hypothesized to have a mean fill of $\mu_1 = 100 + \delta$ ml and a standard deviation of $\sigma_1 = 2 - \sigma_{dev}$ ml, where $\delta > 0$ and $\sigma_{dev} > 0$ are unknown positive constants, what is the minimum value of σ_{dev} such that the probability of falsely concluding that the new process has *increased* variability (i.e., Type I error) is at most 0.01, assuming the mean fill remains at 100 ml (i.e., $\delta = 0$)?

Let X be the amount of liquid dispensed by the machine. X follows a continuous uniform distribution $U(95, 105)$. The mean is $\mu = \frac{95+105}{2} = 100$ ml. The variance is $\sigma^2 = \frac{(105-95)^2}{12} = \frac{100}{12} = \frac{25}{3}$. The standard deviation is $\sigma = \sqrt{\frac{25}{3}} = \frac{5}{\sqrt{3}} \approx 2.887$ ml. The problem statement implies the initial standard deviation is $\sigma_0 = 2$ ml, which contradicts the calculated standard deviation for a uniform distribution from 95 to 105. This suggests a misunderstanding in the problem phrasing or that the uniform distribution is a simplified model, and the actual process has a standard deviation of $\sigma_0 = 2$. We will proceed assuming the standard deviation of the current process is $\sigma_0 = 2$ ml. Final Answer: Approximately 1.37

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18 **[Challenging]** • **Target: Resistant vs. non-resistant statistics** • **Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem**

A set of daily high temperatures in Celsius has mean $\bar{x} = 25.0^\circ\text{C}$ and standard deviation $s = 4.0^\circ\text{C}$. Convert the temperatures to Fahrenheit using $F = 1.8C + 32$. Find the new mean and standard deviation in Fahrenheit.

Mean: Affected by both multiplication and addition: $\bar{x}_F = 1.8(25.0) + 32 = 45.0 + 32 = \mathbf{77.0}^\circ\text{F}$.
Standard Deviation: Affected ONLY by multiplication, not by the addition of 32:
$s_F = 1.8(s_C) = 1.8(4.0) = \mathbf{7.2}^\circ\text{F}$.

19 **[Challenging]** • **Target: Choosing appropriate summary statistics** • **Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing**

A pharmaceutical company is conducting a clinical trial to test the efficacy of a new drug designed to lower blood pressure. They recruit 200 participants and randomly assign them to either the treatment group (receiving the new drug) or the control group (receiving a placebo). After 12 weeks, they measure the reduction in systolic blood pressure for each participant. The company wants to perform a hypothesis test to determine if there is a significant difference in blood pressure reduction between the two groups. The trial is designed with 20 participants in the treatment group and 180 participants in the control group. If the standard deviation of blood pressure reduction in the treatment group is $\sigma_1 = 15$ mmHg and in the control group is $\sigma_2 = 12$ mmHg, and they are planning to use a two-sample t-test, what is the calculated pooled standard deviation s_p if equal variances are assumed, given the following summary statistics for blood pressure reduction (in mmHg): $\bar{x}_1 = 10$ mmHg for the treatment group and $\bar{x}_2 = 7$ mmHg for the control group?

The problem asks for the pooled standard deviation s_p when assuming equal variances for a two-sample t-test. The formula for the pooled variance s_p^2 is given by: $s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$. In this problem, we have: - Sample size for treatment group, $n_1 = 20$ - Sample standard deviation for treatment group, $s_1 = 15$ mmHg - Sample size for control group, $n_2 = 180$ - Sample standard deviation for control group, $s_2 = 12$ mmHg - Total degrees of freedom, $df = n_1 + n_2 - 2 = 20 + 180 - 2 = 198$. Plugging these values into the formula for pooled variance: $s_p^2 = \frac{(20 - 1)(15^2) + (180 - 1)(12^2)}{20 + 180 - 2}$ $s_p^2 = \frac{19(225) + 179(144)}{198}$ $s_p^2 = \frac{4275 + 25776}{198}$ $s_p^2 = \frac{30051}{198}$. The pooled standard deviation is the square root of the pooled variance: $s_p = \sqrt{\frac{30051}{198}}$. Let's examine the choices: We can approximate the value to confirm: $s_p \approx \sqrt{\frac{30051}{198}} \approx \sqrt{151.77} \approx 12.32$. Final Answer: $\sqrt{\frac{20(15^2) + 180(12^2)}{200}}$

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20 [Challenging] • Target: Computing summary statistics from a frequency table • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

A marketing research firm is designing an experiment to test the effectiveness of three different advertising slogans (Slogan A, Slogan B, Slogan C) on product purchase intent. They recruit 240 participants and randomly assign them to one of the three slogan groups. After exposure to their assigned slogan, participants rate their purchase intent on a scale of 1 to 10. The experiment is designed as a completely randomized design. The data collected will be analyzed using ANOVA to compare the mean purchase intent scores across the three groups. In order to determine the minimum number of participants needed in each group for a power analysis, the researchers need to estimate the effect size. A pilot study suggests that the standard deviation of purchase intent scores within each group is approximately $\sigma = 1.8$. They aim for a minimum of $n = 80$ participants per group to achieve the desired statistical power. If the total number of participants available for this experiment were $N = 240$ and they decided to allocate the participants into $k = 3$ treatment groups, with an equal number of participants in each group, what would be the number of participants per group, n? Additionally, if the experimenters decided to use a blocked design instead, where participants are first blocked by age (Young, Middle-aged, Senior) and then randomly assigned slogans within each block, how many blocks would be needed?

The problem asks for two pieces of information: the number of participants per group in a completely randomized design and the number of blocks in a blocked design. Part 1: Participants per group in a completely randomized design. Total number of participants $N = 240$. Number of treatment groups $k = 3$. Since the participants are to be equally divided among the groups, the number of participants per group n is calculated as: $n = \frac{N}{k} = \frac{240}{3} = 80$. Part 2: Number of blocks in a blocked design. The problem states that participants are blocked by age, with three categories: Young, Middle-aged, and Senior. The number of blocks is equal to the number of levels in the blocking variable. Therefore, there are 3 blocks. Combining both parts, the experiment would have 80 participants per group and would require 3 blocks if a blocked design were used. Final Answer: 80 participants per group; 3 blocks

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Mean and median

Find the mean of the data set: $17, 28, 12, 6, 13$.

a. $\frac{81}{5}$

b. $\frac{71}{5}$

c. 19

d. $\frac{76}{5}$ ✓

Mean $= \frac{17+28+12+6+13}{5} = \frac{76}{5}$.

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22 [Foundational] • Target: Range, IQR, and quartiles

Find the mean of the data set: $1, 8, 15, 8, 7$.

a. $\frac{39}{5}$ ✓

b. $\frac{44}{5}$

c. $\frac{39}{4}$

d. $\frac{34}{5}$

Mean $= \frac{1+8+15+8+7}{5} = \frac{39}{5}$.

23 [Foundational] • Target: Standard deviation and variance

Find the mean of the data set: $15, 26, 21, 10, 14$.

a. $\frac{43}{2}$

b. $\frac{91}{5}$

c. $\frac{86}{5}$ ✓

d. $\frac{81}{5}$

Mean $= \frac{15+26+21+10+14}{5} = \frac{86}{5}$.

24 [Foundational] • Target: Visualizing spread with a normal-shaped distribution

Find the mean of the data set: $15, 23, 24, 25, 9$.

a. $\frac{101}{5}$

b. $\frac{91}{5}$

c. $\frac{96}{5}$ ✓

d. 24

Mean $= \frac{15+23+24+25+9}{5} = \frac{96}{5}$.

25 [Foundational] • Target: Comparing distributions using measures of spread

Find the mean of the data set: $22, 24, 23, 15, 27$.

a. $\frac{106}{5}$

b. $\frac{111}{4}$

c. $\frac{111}{5}$ ✓

d. $\frac{116}{5}$

Mean $= \frac{22+24+23+15+27}{5} = \frac{111}{5}$.

26 [Foundational] • Target: Resistant vs. non-resistant statistics

Find the mean of the data set: $26, 7, 1, 1, 20$.

a. 11 ✓

b. 12

c. $\frac{55}{4}$

d. 10

Mean $= \frac{26+7+1+1+20}{5} = 11$.

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27 [Foundational] • Target: Choosing appropriate summary statistics

Find the mean of the data set: $12, 15, 25, 7, 25$.

a. $\frac{89}{5}$

b. 21

c. $\frac{84}{5}$ ✓

d. $\frac{79}{5}$

Mean $= \frac{12+15+25+7+25}{5} = \frac{84}{5}$.

28 [Foundational] • Target: Computing summary statistics from a frequency table

Find the median of the data set: $16, 15, 5, 25, 29, 12$.

a. $\frac{31}{2}$ ✓

b. 17

c. $\frac{33}{2}$

d. $\frac{37}{2}$

Ordered: $5, 12, 15, 16, 25, 29$. Median (average of the two middle values 15 and 16): $\frac{15+16}{2} = \frac{31}{2}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Mean, Median, and Skewness

Find the median of the data set: $31, 29, 29, 23, 23, 36$.

a. 30

b. 27

c. $\frac{57}{2}$

d. 29 ✓

Ordered: $23, 23, 29, 29, 31, 36$. Median (average of the two middle values 29 and 29): $\frac{29+29}{2} = 29$.

30 [Intermediate] • Target: The 1.5 x IQR Rule for Outliers

Find the median of the data set: $12, 30, 16, 1, 22, 37$.

a. 21

b. 19 ✓

c. $\frac{59}{3}$

d. 20

Ordered: $1, 12, 16, 22, 30, 37$. Median (average of the two middle values 16 and 22): $\frac{16+22}{2} = 19$.

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31 [Intermediate] • Target: Standard Deviation and Variance

Find the median of the data set: $24, 30, 40, 6, 31, 5$.

a. $\frac{37}{2}$

b. 27 ✓

c. $\frac{68}{3}$

d. 28

Ordered: $5, 6, 24, 30, 31, 40$. Median (average of the two middle values 24 and 30): $\frac{24+30}{2} = 27$.

32 [Intermediate] • Target: Comparing Distributions with Boxplots

Find the median of the data set: $25, 32, 13, 21, 40, 31$.

a. $\frac{53}{2}$

b. 29

c. 27

d. 28 ✓

Ordered: $13, 21, 25, 31, 32, 40$. Median (average of the two middle values 25 and 31): $\frac{25+31}{2} = 28$.

33 [Intermediate] • Target: Mean and median

Find the median of the data set: $10, 13, 40, 18, 9, 14$.

a. $\frac{52}{3}$

b. 14

c. $\frac{27}{2}$ ✓

d. $\frac{29}{2}$

Ordered: $9, 10, 13, 14, 18, 40$. Median (average of the two middle values 13 and 14): $\frac{13+14}{2} = \frac{27}{2}$.

34 [Intermediate] • Target: Range, IQR, and quartiles

Find the median of the data set: $22, 12, 30, 18, 11, 28$.

a. $\frac{121}{6}$

b. $\frac{39}{2}$

c. 21

d. 20 ✓

Ordered: $11, 12, 18, 22, 28, 30$. Median (average of the two middle values 18 and 22): $\frac{18+22}{2} = 20$.

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35 [Intermediate] • Target: Standard deviation and variance

Find the IQR of the ordered data set: $2, 8, 10, 12, 20, 23, 34, 45$.

a. $\frac{39}{2}$ ✓

b. $\frac{41}{2}$

c. $\frac{37}{2}$

d. $\frac{57}{2}$

Lower half $2, 8, 10, 12$: $Q_1 = \frac{8+10}{2} = 9$. Upper half $20, 23, 34, 45$: $Q_3 = \frac{23+34}{2} = \frac{57}{2}$. IQR $= Q_3 - Q_1 = \frac{39}{2}$.

36 [Intermediate] • Target: Visualizing spread with a normal-shaped distribution

Find the IQR of the ordered data set: $18, 21, 35, 38, 39, 46, 55, 58$.

a. $\frac{43}{2}$

b. $\frac{47}{2}$

c. $\frac{45}{2}$ ✓

d. $\frac{101}{2}$

Lower half $18, 21, 35, 38$: $Q_1 = \frac{21+35}{2} = 28$. Upper half $39, 46, 55, 58$: $Q_3 = \frac{46+55}{2} = \frac{101}{2}$. IQR $= Q_3 - Q_1 = \frac{45}{2}$.

37 [Intermediate] • Target: Comparing distributions using measures of spread

A data set has $Q_1 = 24$ and $Q_3 = 46$. Using the $1.5 \times$ IQR rule, find the upper fence used to detect outliers.

a. 78

b. 68

c. 79 ✓

d. 80

IQR $= Q_3 - Q_1 = 46 - 24 = 22$. Upper fence $= Q_3 + 1.5 \times \text{IQR} = 46 + 1.5(22) = 79$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Resistant vs. non-resistant statistics • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

A data set has $Q_1 = 26$ and $Q_3 = 48$. Using the $1.5 \times$ IQR rule, find the upper fence used to detect outliers.

a. 80

b. 70

c. 81 ✓

d. 82

IQR $= Q_3 - Q_1 = 48 - 26 = 22$. Upper fence $= Q_3 + 1.5 \times \text{IQR} = 48 + 1.5(22) = 81$.

39 [Challenging] • Target: Choosing appropriate summary statistics • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

Find the standard deviation of the data set $29, 22, 31, 38$ (treat as a population), to two decimal places.

a. 6.30

b. 5.40

c. 5.70 ✓

d. 6.00

Mean $= 30$. Deviations: $-1, -8, 1, 8$. Squared deviations: $1, 64, 1, 64$, sum $= 130$. Variance $= \frac{130}{4} = \frac{65}{2}$. Standard deviation $= \sqrt{\frac{65}{2}} \approx 5.70$.

40 [Challenging] • Target: Computing summary statistics from a frequency table • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

Find the standard deviation of the data set $22, 20, 16, 18$ (treat as a population), to two decimal places.

a. 2.54

b. 2.24 ✓

c. 2.84

d. 1.94

Mean $= 19$. Deviations: $3, 1, -3, -1$. Squared deviations: $9, 1, 9, 1$, sum $= 20$. Variance $= \frac{20}{4} = 5$. Standard deviation $= \sqrt{5} \approx 2.24$.

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41 [Challenging] • Target: Mean, Median, and Skewness • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

In a strongly right-skewed distribution of household incomes, which relationship between the mean and median is typically expected?

a. The mean is greater than the median ✓

b. The mean is less than the median

c. The mean equals the median

d. The relationship cannot be determined without sample size

In right-skewed distributions, unusually large values pull the mean toward the right tail, making the mean greater than the median.

42 [Challenging] • Target: Mean, Median, and Skewness • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

A dataset has values: $3, 5, 7, 8, 9, 11, 15$. If the maximum value 15 is replaced with 50, which statistic will NOT change?

a. Median ✓

b. Mean

c. Range

d. Standard Deviation

The median is the 4th value (8). Changing the extreme value from 15 to 50 leaves the order and the middle value (8) unchanged.

43 [Challenging] • Target: The 1.5 x IQR Rule for Outliers • Cross-Lesson: 2.01 Sampling Methods & Bias

What percentage of observations in any distribution lies strictly between Q_1 and Q_3?

a. Approximately 50% ✓

b. 25%

c. 75%

d. 68%

By definition, Q_1 is at the 25th percentile and Q_3 is at the 75th percentile, so exactly the middle 50% of data lies within the interquartile range.

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44 [Challenging] • Target: Standard Deviation and Variance • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

If every observation in a sample has the exact same value of 14, what is the sample standard deviation?

a. 0 ✓

b. 14

c. 1

d. Undefined

When all observations are identical, there is zero variability, so each deviation from the mean ($14 - 14 = 0$) is zero, resulting in $s = 0$.

45 [Challenging] • Target: Comparing Distributions with Boxplots • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

A boxplot shows a long right whisker, a right-half box wider than the left-half box, and no outliers. What is the shape of the distribution?

a. **Skewed to the right** ✓

b. Skewed to the left

c. Roughly symmetric

d. Uniform

An elongated right whisker and a wider upper half of the box indicate that values in the upper 50% are spread over a wider interval, characteristic of a right-skewed distribution.

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